

Self Study:

1. Alternative for SelectMany

- **Purpose:** Flatten nested collections into a single sequence.
 - **Alternative Approaches:** Nested loops, AddRange, or custom iteration.
 - **Use Case:** When you need a single collection from multiple sub-collections.
-

2. Aggregate

- **Purpose:** Combine all elements of a collection into a single result.
 - **Use Case:** For custom reductions, summarizing data, or combining elements logically.
-

3. Yielding (yield return)

- **Purpose:** Produce elements one at a time during iteration without creating a full collection.
 - **Use Case:** Efficient iteration of large sequences or streaming data.
-

4. Lazy Loading

- **Purpose:** Delay creation or retrieval of data until it is actually needed.

- **Use Case:** Improve performance and reduce memory usage, especially with heavy initialization or database queries.
-

5. Multiple Enumeration

- **Purpose:** Occurs when an IEnumerable is iterated multiple times, which can cause inefficiency or unexpected results.
- **Solution / Use Case:** Cache results in a collection to avoid repeated enumeration, especially for expensive operations.